Prateek Singh

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Professional Summary

AI Engineer and Python Backend Engineer with 1.5+ years of experience building production-ready AI solutions, agentic workflows, RAG pipelines, and scalable REST APIs using FastAPI, LangChain, LangGraph, and Azure Cloud. Experienced in taking AI projects from POC to production and optimizing backend systems.

Experience

Member of Technical Staff | CCTech **Sept25-Present**

Design Validation Tool [Python, RAG, Langchain/Langgraph, MCP, Azure AI Search, FastAPI]

- Developed an AI-powered rule extraction pipeline using Azure AI Document Intelligence to convert engineering documents into structured Markdown and extract design rules, achieving 95.4% accuracy across diverse document formats.
- Engineered production-grade Python/FastAPI backend services for document and rule-processing pipelines, implementing RESTful APIs, asynchronous/background execution, Pydantic validation, dependency injection, authentication/RBAC, centralized exception handling, PostgreSQL data access, and Redis-based caching.
- Engineered a multi-stage deduplication pipeline, leveraging text and image hashing, cryptographic document hashing for integrity checks, and embedding-based similarity with a threshold of 0.75, followed by LLM-driven validation to identify duplicate and contradictory engineering rules, reducing redundant processing efforts and improving data consistency.
- Built the end-to-end design validation workflow, transforming extracted rules into structured specifications, generating an execution plan from the specifications, automating NXOpen code generation through MCP, and executing the generated code with 80% of accuracy in the first iteration to validate engineering models against the extracted design requirements.

ML Intern | Infotact Solutions [Python,Machine Learning,Deep Learning,RAG] **Dec24-Apr25**

- Developed cryptocurrency forecasting models (ARIMA, LSTM, Prophet), a real-time fraud detection system using Kafka and Spark, and an SOP-driven conversational AI application for automated customer support workflows.
- Made a GPT-like web application for answering SOP-related questions using RAG, LangChain, and LangGraph to retrieve relevant information and generate context-aware responses with improved retrieval accuracy and relevance.

Data Science intern | IIT Roorkee [Python,Machine Learning,Clustering] **May24-July24**

- Built a gaze fixation mapping algorithm using the Pupil Core eye tracker and K-means clustering, processing 1,000+ driving scenarios for driver behavior analysis and visual attention patterns across varied real-world driving conditions.
- Processed eye-tracking data to map gaze fixation patterns across driving scenarios, supporting analysis of driver visual attention and identification of visual behavior trends under varying road and traffic conditions for safety assessment.

Personal Projects

OpsPilot [Python, FastAPI, LLMs, AI Agents, Redis] Live Demo **Jan26-Jun26**

- Made an incident investigation platform with a React dashboard and FastAPI to simulate **4 production failure scenarios**, manage incidents through REST APIs, collect diagnostic evidence, track incident details and statuses, and generate structured investigation reports using a separate AI investigation service for streamlined incident analysis.
- Implemented AI-assisted root-cause investigation with configurable OpenAI-compatible LLM integration and a deterministic local fallback, supporting **4 failure categories**: HTTP 5xx spikes, database latency, memory pressure, and deployment regressions, to analyze incident evidence and identify probable causes for investigation and troubleshooting.
- Integrated 3 operational components—PostgreSQL persistence, Docker Compose services, and health monitoring—and human approval for remediation plans, enabling incident records, service health checks, and controlled review of suggested actions, with approval recording only and no automatic infrastructure changes across production environments.

SQL Pilot [Python, FastAPI, LangGraph, PostgreSQL] Live Demo **Jun26-Sept26**

- Deployed a private AI data workspace where users upload 8+ file formats and ask questions in plain English, converting files into queryable per-user DuckDB projects without database setup, manual SQL writing, or schema configuration.
- Designed a secure text-to-SQL workflow with 5+ safeguards including schema-aware generation, SQL validation, read-only execution defaults, project isolation, CSRF protection, and controlled mutation settings to prevent unsafe database operations across user-uploaded datasets, sensitive project data, shared environments, and multi-user workflows.
- Implemented a product-style access system with 1 free query per trial user, in-app access requests, and an admin approval dashboard, turning the project into a usable gated SaaS-style demo rather than a one-off technical prototype.

Skills

Programming Languages: Python, Typescript and SQL.
AI & Agentic AI: Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), AI Agents, Semantic Search, Embeddings, LangChain, LangGraph, Azure OpenAI, Azure AI Document Intelligence, Azure AI Search, Vector Databases (FAISS, Pinecone), Reranking, Prompt Engineering.
Backend & Cloud: Backend Development, REST APIs, FastAPI, Microservices, API Integration, Authentication (JWT/OAuth) & RBAC, Asynchronous Programming (AsyncIO), Pydantic, Caching, Database Design, ORM, Query Optimization, PostgreSQL, Redis, Docker, Azure, CI/CD, Git.
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Debugging.

Education

Bachelors of Technology (B.Tech) | National Institute of Technology, Surat | CGPA: 7.69/10 **2021-2025**

Certifications

- Machine Learning with Python
- SQL (Advanced)